with $\mathbf{X} = (\boldsymbol{q}, \boldsymbol{p})^T$ and demonstrating that the required assumptions hold for (6.47). It is reasonable to consider periodic boundary conditions for the position variable, so $\mathcal{Q} = \mathcal{T}^{\mathrm{N_c}}$ where $\mathcal{T}^{\mathrm{N_c}}$ is an $\mathrm{N_c}$-dimensional torus. The use of periodic boundary conditions considerably simplifies the proofs, while also being more relevant to the practical realities of molecular dynamics simulation. The whole phase space is $\mathcal{D} = \mathcal{T}^{\mathrm{N_c}} \times \mathbb{R}^{\mathrm{N_c}}$.

We also consider the potential energy to be smooth (C^∞) with $U(\boldsymbol{q}) \to \infty$ as $|\boldsymbol{q}| \to \infty$. As we have a smooth potential on a bounded (position) domain, we have

$$1 < U_{\min} \leq U(\boldsymbol{q}) \leq U_{\max} < \infty, \quad \forall \boldsymbol{q} \in \mathcal{Q},$$

for some constants $U_{\min}$ and $U_{\max}$.

For the Lyapunov condition (Assumption 2), we choose the Lyapunov function

$$\phi(\boldsymbol{q}, \boldsymbol{p}) = H^l(\boldsymbol{q}, \boldsymbol{p}) = \left(\frac{1}{2} \|\boldsymbol{p}\|^2 + U(\boldsymbol{q}) \right)^l,$$

for a fixed positive integer l. As the potential energy is bounded below by 1, it is clear that $\phi(\boldsymbol{q}, \boldsymbol{p}) \to \infty$ as $|(\boldsymbol{q}, \boldsymbol{p})| \to \infty$. Computing the generator of Langevin dynamics as

$$\mathcal{L}_{\mathrm{LD}} f = \boldsymbol{p} \cdot \nabla_q f - \nabla U(\boldsymbol{q}) \cdot \nabla_p f - \gamma \boldsymbol{p} \cdot \nabla_p f + \gamma \mathrm{k_B T} \Delta_p \phi,$$

we can see that

$$\boldsymbol{p} \cdot \nabla_q \phi - \nabla U(\boldsymbol{q}) \cdot \nabla_p \phi = \mathcal{L}_H \phi = 0,$$

where $\mathcal{L}_H$ is the Lie derivative of our original Hamiltonian system (which preserves all functions of the Hamiltonian). Hence

$$\mathcal{L}_{\mathrm{LD}} \phi = -\gamma \boldsymbol{p} \cdot \nabla_p \phi + \gamma \mathrm{k_B T} \Delta_p \phi.$$

Using $\|\boldsymbol{p}\|^2 < 2H$, we compute derivatives to find

$$\begin{aligned} \boldsymbol{p} \cdot \nabla_p \phi &= l H^{l-1} \|\boldsymbol{p}\|^2 = 2l\phi - 2l H^{l-1} U(\boldsymbol{q}), \\ \Delta_p \phi &= l(l-1) H^{l-2} \|\boldsymbol{p}\|^2 + \mathrm{N_c} l H^{l-1} \leq l(l + \mathrm{N_c} - 1) H^{l-1}. \end{aligned}$$

As H^l grows faster than H^{l-1}, for any fixed l we are able to find positive constants c and d such that

$$l(l + \mathrm{N_c} - 1) H^{l-1} \leq \frac{l}{2} H^l + c, \quad 2l H^{l-1} U(\boldsymbol{q}) < 2l H^{l-1} U_{\max} \leq \frac{l}{2} H^l + d.$$

giving

$$\begin{aligned}\mathcal{L}_{\mathrm{LD}}\phi &\leq -2\gamma l\phi + 2\gamma l H^{l-1}U(\boldsymbol{q}) + \gamma l(l+\mathrm{N_c}-1)H^{l-1},\\ &\leq -\gamma l\phi + \gamma c + \gamma d,\\ &\leq -\alpha\phi + \delta,\end{aligned}$$

satisfying Assumption 2.

The purpose of using the l exponent in the Lyapunov function is to ensure that we can apply the theorem to a wide enough class of observables, as (6.48) is only valid for functions f with magnitude less than or equal to the Lyapunov function. As we have $\phi \geq U^l_{\min}$ and growing like $\|\boldsymbol{p}\|^{2l}$, we can choose l to be sufficiently large to bound a wide class of observables. However, for sufficiently fast-growing functions (such as exponentals) a different Lyapunov function would be required (if their averages do, in fact, converge).

We note that the condition of Assumption 1(ii) can be shown by demonstrating that *Hörmander's condition* holds over the entire space $\mathcal{D}$. This is a condition on the commutators or Lie brackets of vector fields.

The *Lie bracket* of a pair of vector fields $\boldsymbol{u}(\boldsymbol{z})$, $\boldsymbol{v}(\boldsymbol{z})$ is another vector field

$$[\boldsymbol{u},\boldsymbol{v}] = \boldsymbol{v}'\boldsymbol{u} - \boldsymbol{u}'\boldsymbol{v},$$

where $\boldsymbol{u}'$ is the Jacobian matrix of the vector field $\boldsymbol{u}$. In coordinates, if we let $\boldsymbol{w} = [\boldsymbol{u},\boldsymbol{v}]$, then the coordinates of $\boldsymbol{w}$ are

$$w_i = \sum_{j=1}^{m} \frac{\partial v_i}{\partial z_j}u_j - \frac{\partial u_i}{\partial z_j}v_j.$$

The Lie bracket is a bilinear and skew-symmetric operation:

- $[\alpha\boldsymbol{u} + \beta\boldsymbol{v},\boldsymbol{w}] = \alpha[\boldsymbol{u},\boldsymbol{w}] + \beta[\boldsymbol{v},\boldsymbol{w}]$,
- $[\boldsymbol{u},\boldsymbol{v}] = -[\boldsymbol{v},\boldsymbol{u}]$ and $[\boldsymbol{u},\boldsymbol{u}] = \boldsymbol{0}$.

Definition 6.1 (Hörmander's Condition) Let $\mathcal{U} \subset \mathbb{R}^m$ be open. The SDE (6.43) with C^∞ vector fields $\boldsymbol{b}_0,\boldsymbol{b}_1,\boldsymbol{b}_2,\ldots,\boldsymbol{b}_r : \mathcal{U} \to \mathbb{R}^m$ are said to satisfy Hörmander's condition at $\boldsymbol{z} \in \mathcal{U}$ if the vector space spanned by the collection of iterated Lie brackets

$$\boldsymbol{b}_0,\boldsymbol{b}_1\ldots,\boldsymbol{b}_r,[\boldsymbol{b}_0,\boldsymbol{b}_1],[\boldsymbol{b}_0,\boldsymbol{b}_2],\ldots,[\boldsymbol{b}_{r-1},\boldsymbol{b}_r],[\boldsymbol{b}_0,[\boldsymbol{b}_0,\boldsymbol{b}_1]]\ldots$$

(evaluated at $\boldsymbol{z}$) is $\mathbb{R}^m$.

Computing the iterated Lie brackets becomes, in many cases, an arduous task.